Video Prediction

Computer Vision project by Maiola and Matini

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The task

Video prediction is a complex task that requires to produce the next frames from the initial part of a video sequence.

There are two main branches of research:

- Using pretext tasks (e.g. depth estimation, optical flow…)
- Using only the original grayscale/RGB frames

The datasets

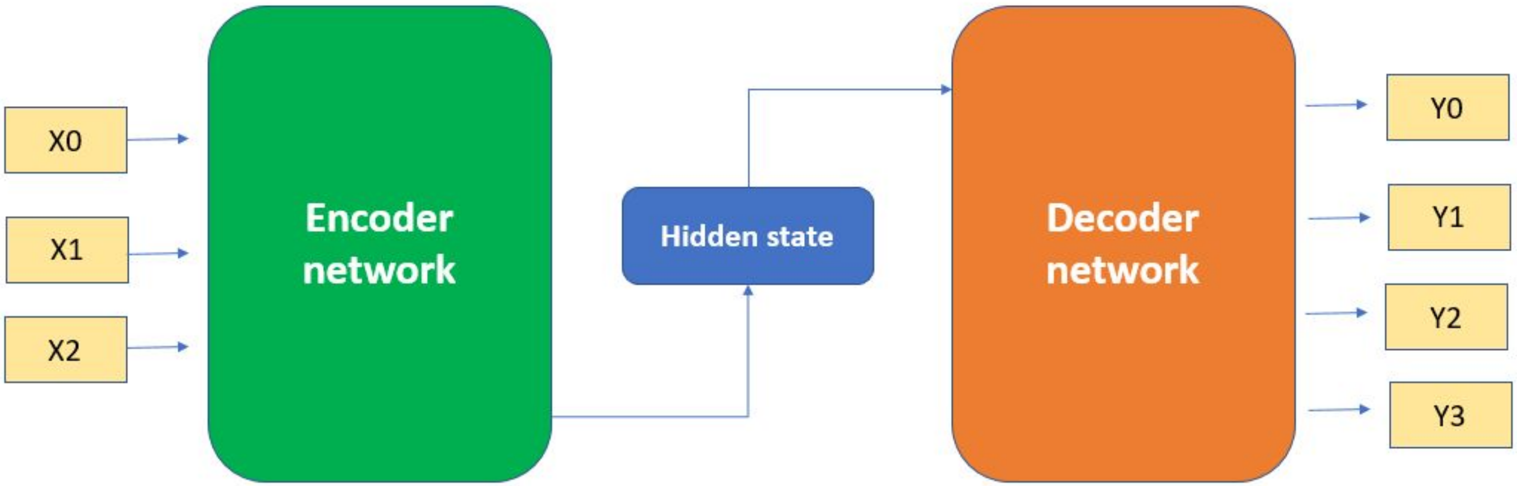
We employed two very different datasets:

- UCF101: a dataset originally produced for action recognition, containing a series of video sequences with 101 real world actions

- MovingMNIST: a synthetic dataset containing videos with two moving numbers

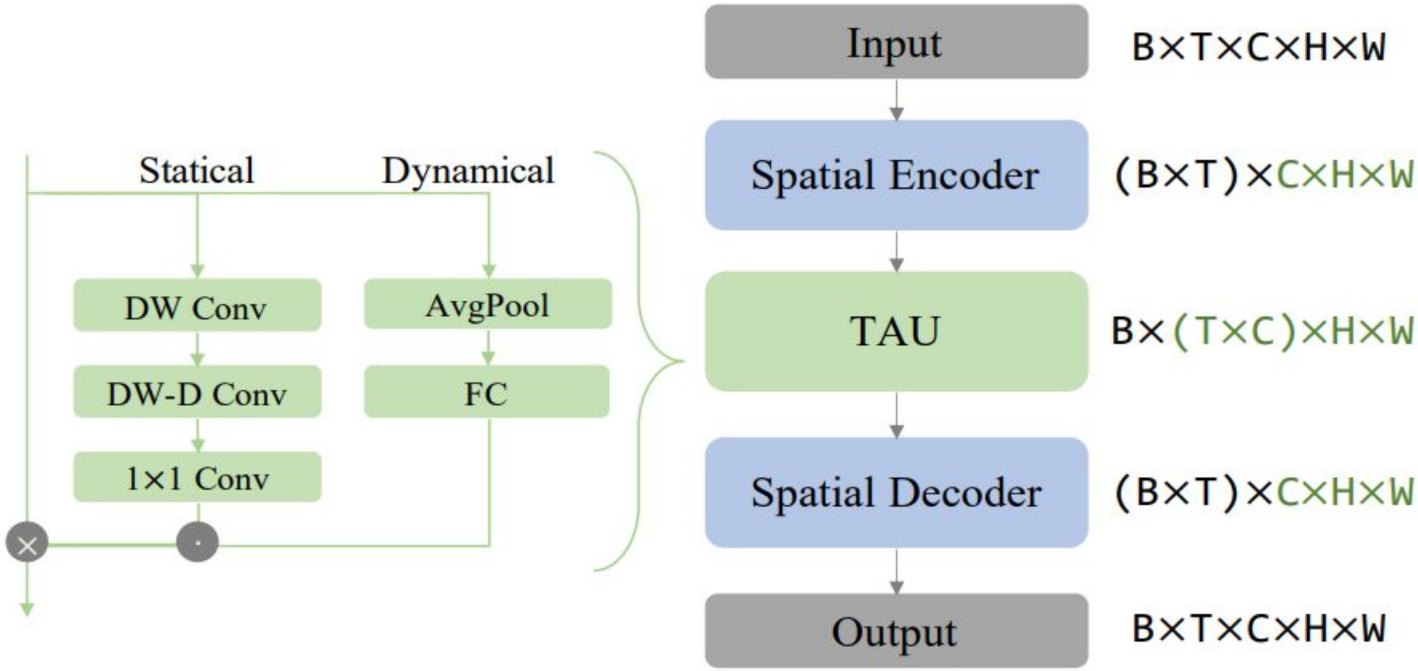
The architecture

Encoder-Decoder, Attention Layer with 3 main components



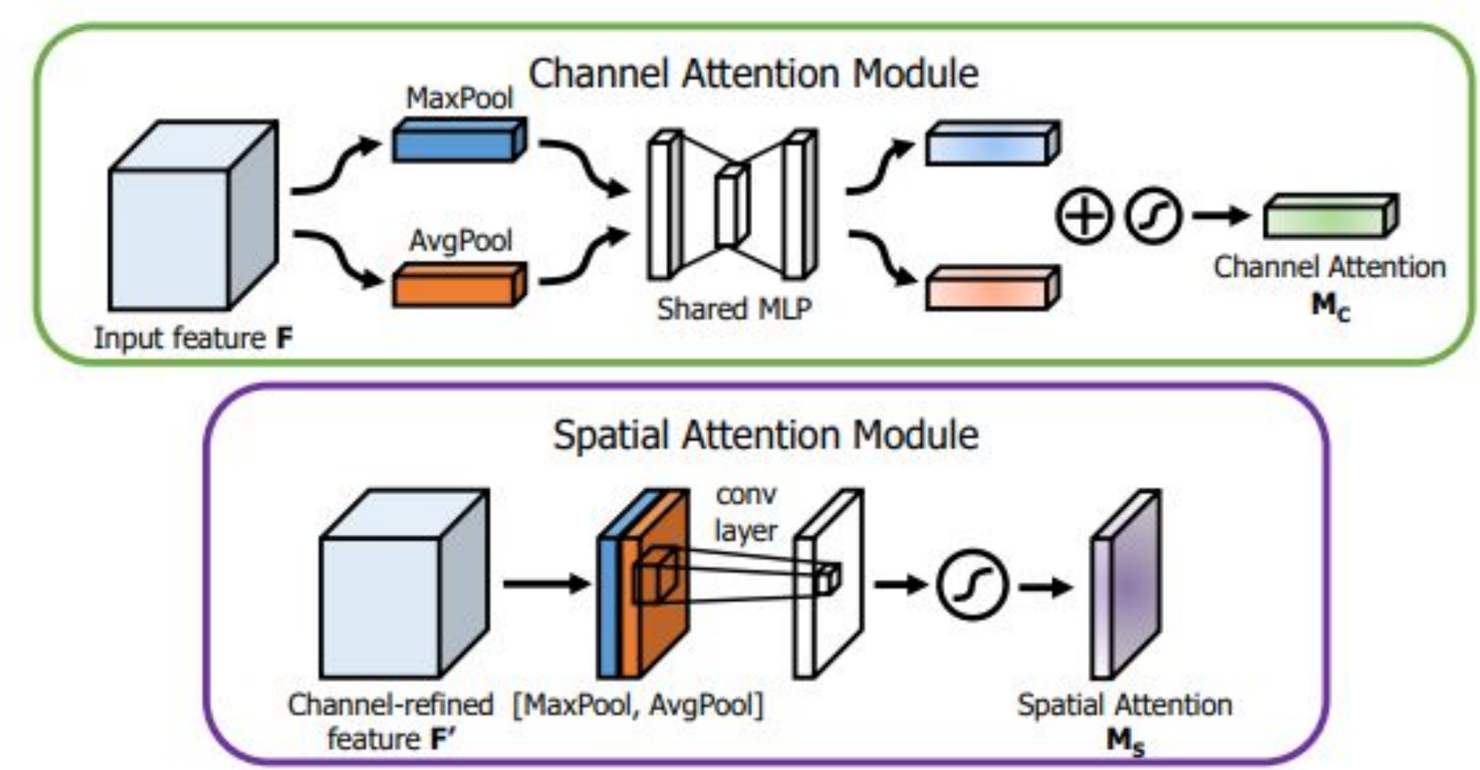
Temporal Attention Unit

Capturing intra-frame and inter-frame dependencies using Statical and Dynamical attention



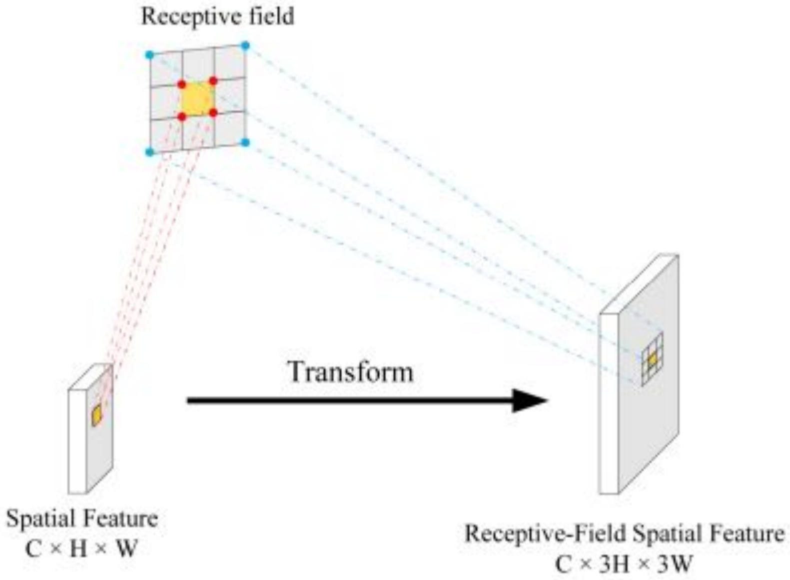
CBAM

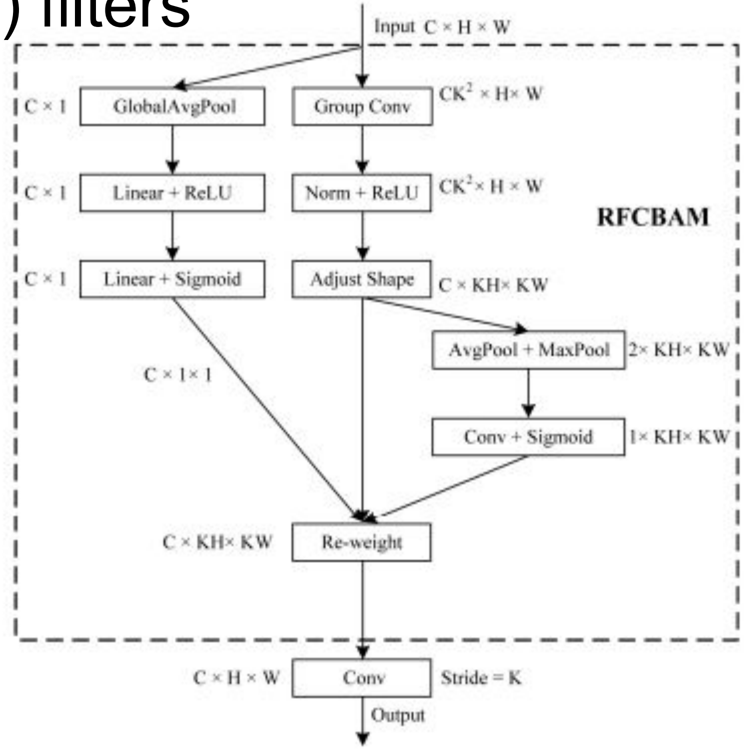
After temporal attention, we thought about employing spatial and channel attention



Receptive Field Convolutions Attention

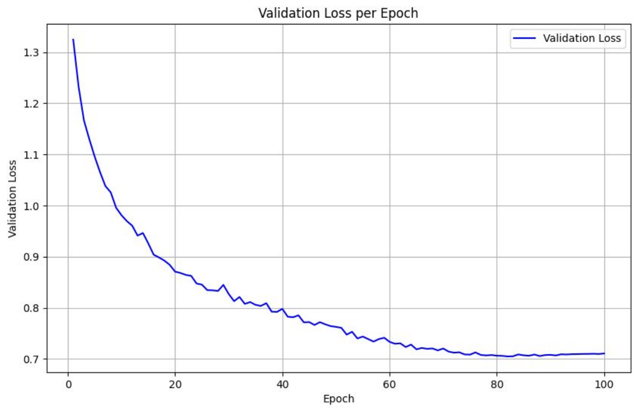
Receptive field convolutions aims to mitigate parameters sharing problem using the entire receptive field to compute attention with large (at least 3x3) filters

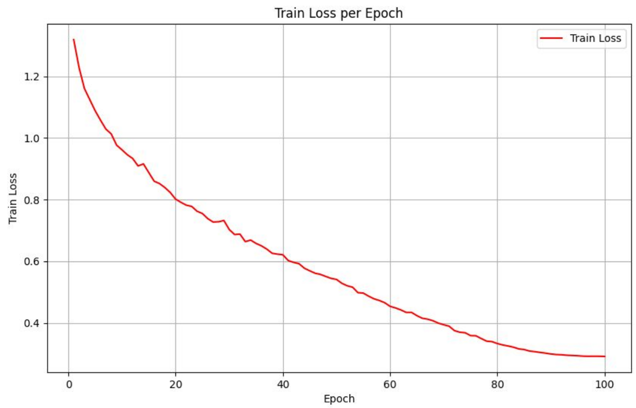




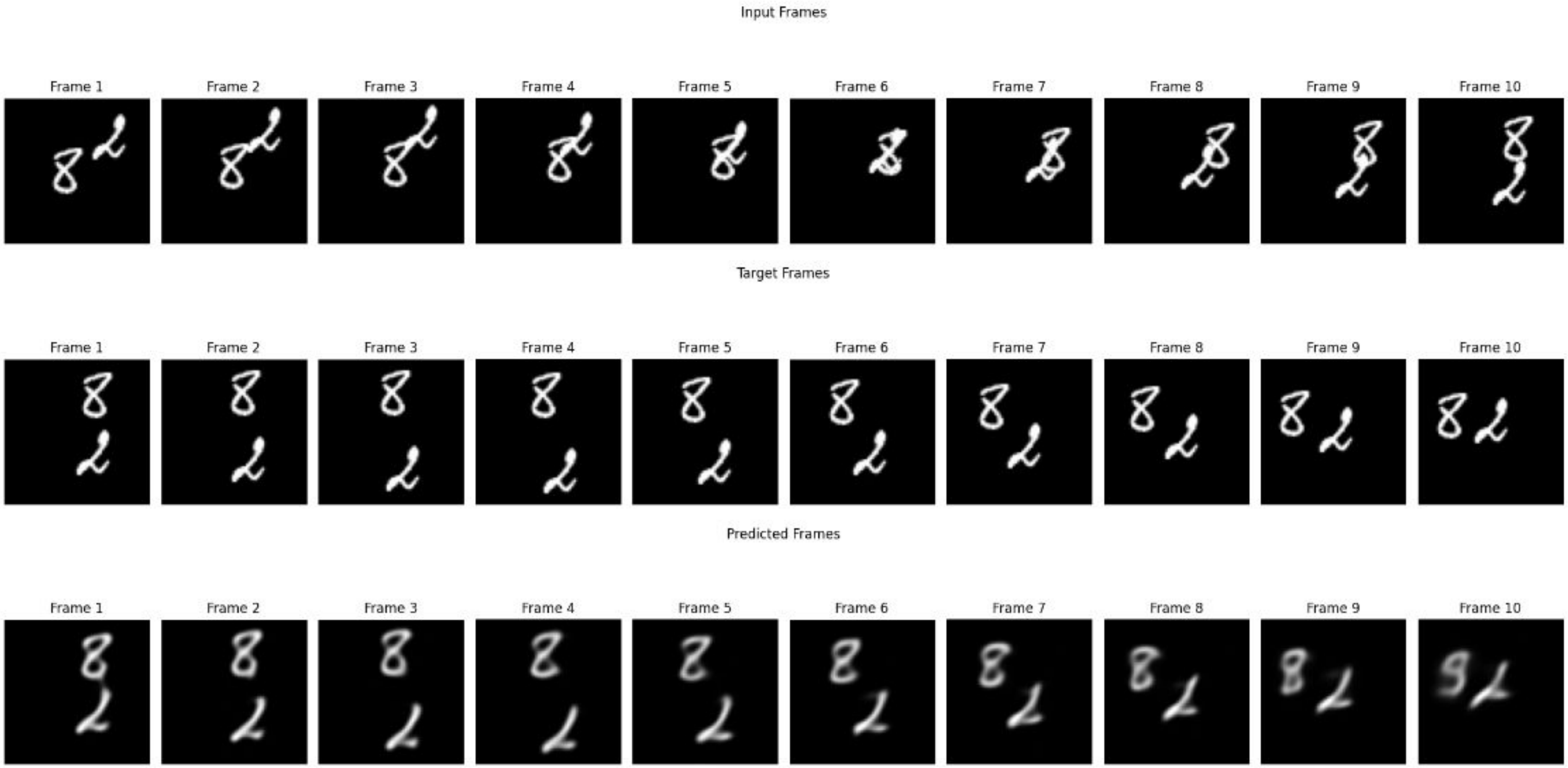
Results

Model Type/Attention Position	0/0	1/0	2/2
MovingMNIST	0,73749	**0,71143**	0,78799
UCF101	0,14455	0,14547	**0,14369**
MovingMNIST after train on UCF101	1,11446	1,0828	**1,0691**
Using only MSE			
MovingMNIST	0,48049	**0,46842**	0,50923
UCF101	0,13981	0,14377	**0,13891**
MovingMNIST after train on UCF101	0,99546	0,96546	**0,9528**





Results


Input Frames
Frame 1
Frame 2
Frame 3
Frame 4
Frame 5
Frame 6
Frame 7
Frame 8
Frame 9
Frame 10
Target Frames
Frame 1
Frame 2
Frame 3
Frame 4
Frame 5
Frame 6
Frame 7
Frame 8
Frame 9
Frame 10
Predicted Frames
Frame 1
Frame 2
Frame 3
Frame 4
Frame 5
Frame 6
Frame 7
Frame 8
Frame 9
Frame 10

Future research objectives

- Optimize the usage of receptive fields convolutions
- Produce more light-weight models
- Use pretext tasks